

Samuel Valiquette

Institution: Département de mathématiques, Université de Sherbrooke, Sherbrooke, J1K 2R1

Position: Assistant Professor in Statistics

E-mail: samuel.valiquette@usherbrooke.ca * *Website:* s-valiquette.github.io

Education

Joint PhD degree in Mathematics/Biostatistics

Université de Montpellier/Sherbrooke

PhD's degree program

August 2020 - August 2024

Thesis title: [Sur les données de comptage dans le cadre des valeurs extrêmes et la modélisation multivariée](#)

Supervisors: Marchand, É., Mortier, F., Peyhardi, J. & Toulemonde, G.

Joint Master degree in Biostatistics

Université de Montpellier/Sherbrooke

Master's degree program

August 2018 - August 2020

Thesis title: [Théorie des valeurs extrêmes dans le cadre des mélanges de Poisson](#)

Supervisors: Marchand, É., Mortier, F., Peyhardi, J. & Toulemonde, G.

Bachelor's degree in Mathematics/Statistics

Université de Montréal

Bachelor's degree program

August 2015 - August 2018

Work experience

Assistant Professor in Statistics

Sherbrooke, Canada

Université de Sherbrooke

Since July 2026

Département de mathématiques

Postdoctoral Fellow in Statistics

Montréal, Canada

McGill University

August 2024 - June 2026

Supervisors : Christian Genest & Johanna Nešlehová

BRP

Sherbrooke, Canada

Intern (Master's student)

January 2019 - May 2019

Project: Time series analysis of recreational vehicles

Employment and Social Development Canada

Gatineau, Canada

Intern (Bachelor's student)

May 2018 - August 2018

Project: Statistical analysis of service quality

Division: Analytical research

Statistics Canada

Ottawa, Canada

Intern (Bachelor's student)

May 2017 - August 2017

Project: Master File for graduate students in Canada

Division: Household Survey Methods

Academic Work Experience

Supervisor

Since January 2025

- Cedric Phillips: Hierarchical copula dependence models ([MATH 470](#), Winter 2026)
- Ziyu Zhou: Multivariate Modeling via Tree Pólya Splitting ([MATH 410](#), Winter 2025)

Teaching

Since August 2023

- [STT 390](#): Mathematical and Inferential Statistics (Winter 27)
- [STT 760](#): Mathematics for Artificial Intelligence (Fall 26, Fall 24, Fall 23)
- [MATH 262](#): Intermediate Calculus (Fall 25)

Teaching Assistant/Corrector

Since August 2018

- [STT 438](#): Computational Statistic (Fall 23, Fall 22)
- [STT 760](#): Mathematics for Artificial Intelligence (Fall 22, Fall 21)
- [STT 563](#): Linear Statistical Model (Winter 22, Winter 21)
- [STT 289](#): Probability (Fall 21)
- [MAT 153](#): Introduction to Linear Algebra (Fall 20)
- [MAT 356](#): Analytical Geometry (Winter 19)
- [CQP 201](#): Linear Algebra and Matrix calculation (Fall 18)

Service and leadership

- Member of the [Bilingualism Committee](#) for the Statistical Society of Canada (2026-2029).
- Co-organizer of the Séminaire de Statistique at Université de Sherbrooke (2026), alongside [Taoufik Bouezmarni](#).
- Co-organizer of the [McGill Statistics Seminar \(2024-2026\)](#), alongside [Mehdi Dagboug](#).
- Reviewer for the [Journal of Multivariate Analysis](#) and [Sankhya A](#).
- Judge for the oral and poster presentation sessions at the [Thirteenth annual Canadian Statistics Student Conference \(CSSC\)](#), May 24, 2025.
- Executive member (Vice-president) of the undergraduate student union at the Département de mathématiques et de statistique of the Université de Montréal for the academic year 2017-2018.

Articles

On the Williamson d-transform

Genest, C., Nešlehová, J. G., Ouimet, F., **Valiquette, S.**

In preparation
August 2026

A class of multivariate Bernoulli distributions generated by an aggregation-tree model

Valiquette, S., Genest, C., Nešlehová, J. G.

In revision
August 2026

On conditional independence within multivariate Bernoulli random vectors

Valiquette, S., Genest, C.

Statistics & Probability Letters
July 16, 2026

Tree Pólya Splitting distributions for multivariate count data

Valiquette, S., Peyhardi, J., Marchand, É., Toulemonde, G., Mortier, F.

Journal of Multivariate Analysis
September 12, 2025

Asymptotic tail properties of Poisson mixture distributions

Valiquette, S., Toulemonde, G., Peyhardi, J., Marchand, É., Mortier, F.

Stat
September 26, 2023

Talks & Workshops

A Class of Multivariate Bernoulli Distributions Generated by an Aggregation-Tree Model <i>2026 SSC Annual Meeting</i>	<i>Hamilton, Canada</i> <i>June 2, 2026</i>
YES - Dependence Modeling <i>Eurandom (Participant)</i>	<i>Eindhoven, Netherlands</i> <i>June 30, 2025</i>
Asymptotic Tail Properties of Poisson Mixture Distributions <i>2025 SSC Annual Meeting</i>	<i>Saskatoon, Canada</i> <i>May 26, 2025</i>
Modélisation multivariée par agrégation hiérarchique <i>Séminaires de statistique (Invited)</i>	<i>Sherbrooke, Canada</i> <i>March 6, 2025</i>
Modèle multivarié discret Tree Pólya Splitting <i>Séminaires de STATQAM (Invited)</i>	<i>Montréal, Canada</i> <i>January 30, 2025</i>
Tree Pólya Splitting distributions for multivariate count data <i>McGill Statistics Seminars (Invited)</i>	<i>Montréal, Canada</i> <i>January 17, 2025</i>
Comportement extrême des mélanges Poisson <i>Séminaire du CIMMUL (Invited)</i>	<i>Québec, Canada</i> <i>September 27, 2024</i>
Modèle multivarié discret Tree Pólya Splitting <i>Rencontres scientifiques Montpellier Sherbrooke (Invited)</i>	<i>Sherbrooke, Canada</i> <i>June 20, 2024</i>
Multivariate discrete Tree Pólya Splitting distributions <i>2024 SSC Annual Meeting</i>	<i>Saint John's, Canada</i> <i>June 4, 2024</i>
Comportement asymptotique des mélanges Poisson <i>Semaine des études supérieures - Université de Sherbrooke (Poster)</i>	<i>Sherbrooke, Canada</i> <i>January 24, 2024</i>
Asymptotic tail properties of Poisson mixture distributions <i>2023 Joint Statistical Meetings</i>	<i>Toronto, Canada</i> <i>August 10, 2023</i>
Extreme value theory for the choice of mixture Poisson models <i>52èmes Journées de Statistique de la Société Française de Statistique (Virtual)</i>	<i>Nice, France</i> <i>June 10, 2021</i>
Le Bon, le Standard et le Bayésien <i>Club Mathématique de Sherbrooke</i>	<i>Sherbrooke, Canada</i> <i>January 24, 2019</i>
Le Bon, le Standard et l'Exact <i>Séminaires Universitaire en Mathématiques à Montréal</i>	<i>Montréal, Canada</i> <i>January 12, 2019</i>

Science Popularization

Publications

- Genest, C., Valiquette, S. (In preparation). Les mathématiques du Lotto 6/49. Bulletin AMQ.
- Genest, C., Valiquette, S. (2026). Le Lotto 6/49. Accromath, 21 (2), in press.

Presentations

- Presentations of graduate studies at François-Williams highschool (April 10, 2024).

A panel of PhD students from various fields presented their perspectives on university life and the benefits of pursuing advanced degrees to high school students. My aim was to make the work of a statistician and the value of higher education in mathematics accessible and engaging.

Scholarships & Grants

Starting Grant <i>Université de Sherbrooke - Faculté des sciences</i>	<i>2026/08 - 2027/08</i> <i>30 000\$</i>
ISM Travel Grant <i>Institut des Sciences Mathématiques</i>	<i>2024/06 - 2024/06</i> <i>500\$</i>
ISM Graduate Scholarships 2023-2024 <i>Institut des Sciences Mathématiques</i>	<i>2023/08 - 2024/08</i> <i>5 000\$</i>
Valorisation des travaux des étudiantes et des étudiants 2022-2023 <i>Université de Sherbrooke - Faculté des sciences</i>	<i>2023/08 - 2024/08</i> <i>1 000\$</i>
ISM Graduate Scholarships 2019-2020 <i>Institut des Sciences Mathématiques</i>	<i>2019/08 - 2020/08</i> <i>4 000\$</i>
Bourse Frontenac 2019-2020 <i>Fonds de recherche du Québec - Nature et technologies</i>	<i>2019/08 - 2020/08</i> <i>6 000\$</i>

Technical skills

Programming Languages/Tools	C, C++, Javascript, R, Python, L ^A T _E X
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Language proficiencies

French	Native language
English	Fluent

Research Interests

Multivariate models
Dependence structure
Extreme value theory
Copulas
Discrete distribution
Hierarchical models
Probabilistic graphical model
Inference
Stochastic process